



Human-in-the-Loop AI for IT Ticket Triage & Resolution

AI proposes · Humans decide · The system learns — and proves it

IT Ticket Manager — AI-Powered Ticket System · Product demo & presenter guide

How to run this session

Agenda (~20 min) — Story (4 min) → Live walkthrough (12 min) → Metrics & value (4 min) → Q&A.

Before you start (setup checklist):

- Start the frontend dev server, open <http://localhost:5173> and sign in.
- Toggle the sidebar switch to **Demo** (orange dot) → the whole app runs on seeded data, **no backend needed**.
- Have two tabs ready: the main app, and **Ticket Evolution** (it opens in a new tab).
- Zoom the browser to ~110–125% so the room can read the UI.

This deck doubles as your script — **speaker notes on every slide** contain what to say and what to click.

The problem we're solving

- IT service desks are **flooded** with tickets — triage is slow, manual and repetitive.
- Routing to the **wrong team** and writing **first replies** from scratch waste hours every day.
- Teams don't trust **black-box AI** they can't understand or correct.
- Leaders lack the **evidence** to know whether automation is actually working.

The goal: cut manual effort and resolution time **without** handing control to an opaque model.

Our answer in one line

The AI drafts the triage and the reply. A human stays in control. Every decision teaches the model — and everything is measured.

1 · Classify & Resolve

Auto-triage + AI-drafted replies

2 · Explain

Show *why*, not just *what*

3 · Learn

Human feedback improves the model

4 · Prove

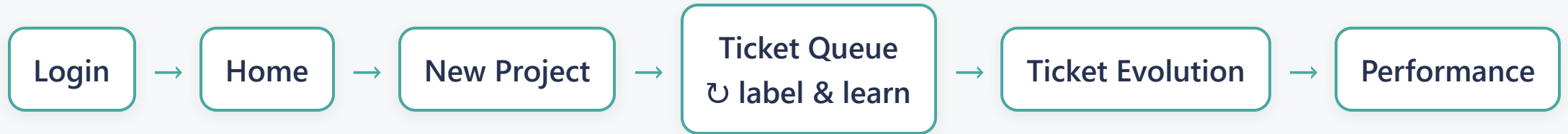
Benchmark & measure impact

Who uses it

Persona	Goal	Where in the app
Ticket Labeler / Agent	Confirm or correct AI triage, fast	Ticket Queue
Support Operator (Tier 2)	Send a quality first reply in seconds	Ticket Evolution
Team Lead / Analyst	Track accuracy, effort saved & KPIs	Performance

One human-in-the-loop workflow, three points of value.

The journey we'll walk through



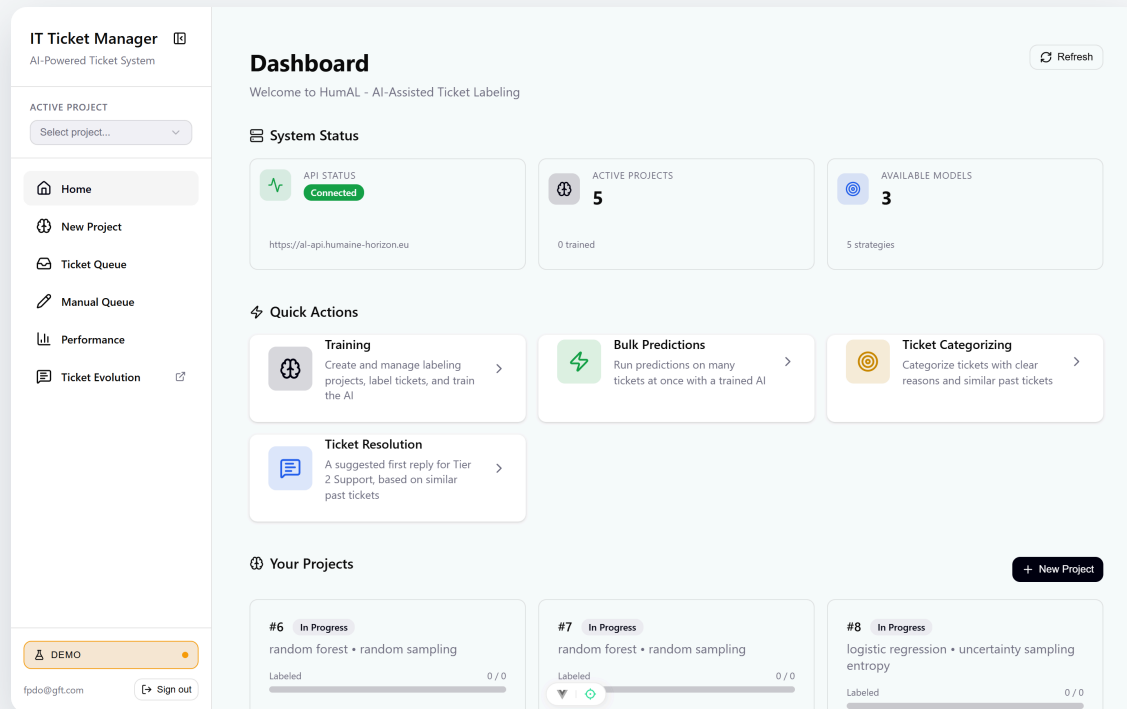
The rest of the deck follows this exact path — each screen with the idea behind it, then a close-up of its inputs and buttons.

The flow, step by step

Each screen — the concept, then the controls

Step 1 · Home

Sign in → the operator's home base



Flow: A secure JWT login opens the **dashboard** — API status, active projects and available AI models — with **quick actions** into every workflow.

Step 1 · Home — the controls

What's on the dashboard

Element	What it tells you / does
System Status → API Status	Backend health: <i>Connected</i> / <i>Checking...</i> / <i>Disconnected</i>
Active Projects · <i>X trained</i>	How many labeling projects exist, how many are trained
Available Models · <i>X strategies</i>	AI models and labeling strategies ready to use
Quick Actions	Shortcut cards: <i>Training</i> , <i>Bulk Predictions</i> , <i>Ticket Categorizing</i> , <i>Ticket Resolution</i>
Your Projects → Open	Jump into an existing project; badge shows <i>Trained</i> / <i>In Progress</i>
New Project / Refresh	Start a new project · re-check status

Step 2 · New Project

Choose the AI — *and how it learns*

The screenshot shows the 'New Project' form in the 'IT Ticket Manager' application. The sidebar on the left contains navigation links: Home, New Project (active), Ticket Queue, Manual Queue, Performance, and Ticket Evolution. The main content area is titled 'New Project' with a subtitle 'Set up a new labeling project. You'll go to the queue once it is ready.' The form itself is titled 'Create Project' and includes instructions: 'Pick an AI model, a labeling strategy, and the categories to label.' It features several input fields: 'AI Model' (a dropdown menu), 'Labeling Strategy' (a dropdown menu), 'Training Data (CSV)' (a file selection button showing 'al_demo_train_data.csv'), and 'Test Data (CSV)' (a file selection button showing 'al_demo_test_data.csv'). Below these is a note: 'Uploading files from your machine is not available yet — disabled for demo purposes.' The 'Categories (Teams)' section has a 'Load from Data' button and a text input field with the placeholder 'Click "Load from Data" or add categories manually below'. Below this is another text input field for 'Type a team name...' with an '+ Add' button. At the bottom of the form is a '+ Create Project' button. The footer of the application shows a 'DEMO' label, the email 'fpdo@gift.com', and a 'Sign out' button.

- Pick an **AI Model** and the **Categories (Teams)** to route to.
- Choose a **Labeling Strategy** — *uncertainty, entropy, margin, query-by-committee, random* — deciding which tickets the AI asks a human about first.

This is **active learning**: spend human effort on the most informative tickets.

Step 2 · New Project — the controls

Every input & button

Control	What it does
AI Model (dropdown)	The ML algorithm that learns to route tickets
Labeling Strategy (dropdown)	Picks <i>which</i> tickets to ask the human about first (active learning)
Training / Test Data (CSV)	Data source (<i>fixed in the demo; upload disabled</i>)
Download CSV Template	Gives the exact columns: <code>title_anon, description_anon, service_name, ...</code>
Categories (Teams) → Load from Data	Auto-fills the routing categories from the dataset
Type a team name... → Add / chip ✕	Add or remove categories manually
Create Project	Creates the instance and opens the Ticket Queue (<i>needs model + strategy + categories</i>)

Step 3 · Ticket Queue

AI triage — *Classification*

The screenshot displays the 'IT Ticket Manager' interface, specifically the 'Ticket Queue' view. The left sidebar contains navigation links: Home, New Project, Ticket Queue (selected), Manual Queue, Performance, and Ticket Evolution. The main content area shows a ticket titled 'Cannot access network drive Z:' with a description: 'I am getting "network path not found" when trying to access the Z: drive. Other colleagues can access it fine.' The AI suggestion is 'Network Team' with a 79% confidence score. Below this, a bar chart shows the top 3 suggestions: Network Team (78.7%), Help Desk L2 (12.3%), and Desktop Support (3.3%). The interface also includes a 'Skip this ticket' section with options like 'I'm Tired', 'Difficult Ticket', and 'I Don't Know'. A 'Why this suggestion' section compares the current ticket with a 'CLOSEST PAST TICKET' (TKT-10029-N1) and a 'Previously reported' ticket. The bottom section shows 'Similar Tickets by Category'.

IT Ticket Manager
AI-Powered Ticket System

ACTIVE PROJECT
Select project...

Home
New Project
Ticket Queue
Manual Queue
Performance
Ticket Evolution

DEMO

fpdo@gft.com Sign out

Ticket Queue

TKT-10029 · Security · 21 Aug 2026, 09:10

Cannot access network drive Z:

I am getting "network path not found" when trying to access the Z: drive. Other colleagues can access it fine.

AI SUGGESTION **Network Team** 79% Confirm

TOP 3 SUGGESTIONS

Suggestion	Confidence Score
Network Team	78.7%
Help Desk L2	12.3%
Desktop Support	3.3%

Reassign to... Reassign

Skip this ticket:
I'm Tired Difficult Ticket I Don't Know

Why this suggestion

CURRENT TICKET
Cannot access network drive Z:
I am getting "network path not found" when trying to access the Z: drive.
Other colleagues can access it fine.

CLOSEST PAST TICKET TKT-10029-N1 Network Team 74% match
Previously reported — Cannot access network drive Z:
I am getting "network path not found" when trying to access the Z: drive.
Other colleagues can access it fine. Resolved by reassigning to the responsible team.

Similar Tickets by Category
The closest past ticket for each of the AI's top suggestions.

The AI answers **two questions at once**, each with a **confidence score**: what kind of ticket (13+ types) and which team handles it (17 teams). Here: **Application Support · 84%**, with a top-3 breakdown.

Powered by sentence embeddings + fine-tuned language models — no hand-written rules.

Step 3 · Ticket Queue — the controls

Find, decide, move on

Control	What it does
Search / Filters / Sort	Find tickets; sort by <i>Newest</i> or <i>Certainty (High→Low)</i>
Get AI Suggestion	Runs the model on the selected ticket
Confirm ()	Accept the AI's team as the label
Reassign to... → Reassign ( - )	Override with the correct team
Skip: <i>I'm Tired</i> · <i>Difficult Ticket</i> · <i>I Don't Know</i>	Labeled feedback instead of a guess
Bulk select () · Select all ( + )	Label many tickets at once
Break modal	Nudges a pause — " <i>quality matters more than speed</i> "

Step 3 · detail

"Why this suggestion?" — Explainability

The screenshot displays the 'IT Ticket Manager' interface, an AI-Powered Ticket System. The left sidebar contains navigation links: Home, New Project, Ticket Queue (selected), Manual Queue, Performance, and Ticket Evolution. The main area shows a 'Ticket Queue' with a list of tickets on the left and a detailed view of ticket TKT-10029 on the right. The ticket title is 'Cannot access network drive Z:' and it was created on 21 Aug 2026 at 09:10. The description states: 'I am getting "network path not found" when trying to access the Z: drive. Other colleagues can access it fine.' An AI suggestion from the 'Network Team' is shown with a 79% confidence score and a 'Confirm' button. Below this, a 'TOP 3 SUGGESTIONS' bar chart lists: Network Team (78.7%), Help Desk L2 (12.3%), and Desktop Support (3.3%). At the bottom, a 'Why this suggestion' section compares the current ticket with a 'CLOSEST PAST TICKET' (TKT-10029-N1) from the 'Network Team' with a 74% match. The comparison highlights word-by-word weights, showing that words like 'network', 'path', 'not', 'found', 'trying', 'access', 'Z:', 'drive', 'colleagues', and 'access' are key in both tickets. A 'Similar Tickets by Category' section at the bottom shows the closest past tickets for each of the AI's top suggestions.

IT Ticket Manager
AI-Powered Ticket System

ACTIVE PROJECT
Select project...

Home
New Project
Ticket Queue
Manual Queue
Performance
Ticket Evolution

DEMO
Sign out

Ticket Queue

TKT-10029 · Security · 21 Aug 2026, 09:10

Cannot access network drive Z:

I am getting "network path not found" when trying to access the Z: drive. Other colleagues can access it fine.

AI SUGGESTION: Network Team 79% Confirm

TOP 3 SUGGESTIONS

Suggestion	Match
Network Team	78.7%
Help Desk L2	12.3%
Desktop Support	3.3%

Reassign to... Reassign

Skip this ticket:
I'm Tired Difficult Ticket I Don't Know

Why this suggestion

CURRENT TICKET
Cannot access network drive Z:
I am getting "network path not found" when trying to access the Z: drive.
Other colleagues can access it fine.

CLOSEST PAST TICKET: TKT-10029-N1 Network Team 74% match
Previously reported — Cannot access network drive Z:
I am getting "network path not found" when trying to access the Z: drive.
Other colleagues can access it fine. Resolved by reassigning to the responsible team.

Similar Tickets by Category
The closest past ticket for each of the AI's top suggestions.

- Highlighted words show what supports vs. opposes the prediction.
- The closest past ticket (86% match) gives the agent concrete evidence.
- Toggle Show / Hide Details to expand the word-by-word weights.

No black box — explanations build trust, speed decisions, and help catch mistakes.

Power-user tip

Keyboard shortcuts — label at speed

Key	Action	Key	Action
i / ↓	Next ticket	c	Confirm prediction
k / ↑	Previous ticket	1–9	Assign to team 1–9
x	Toggle bulk select	Shift + a	Select all visible
Esc	Close / clear	?	Show shortcuts

A trained labeler never leaves the keyboard — read, decide, next.

Step 4 · Confirm or correct

The learning loop — *Human feedback*

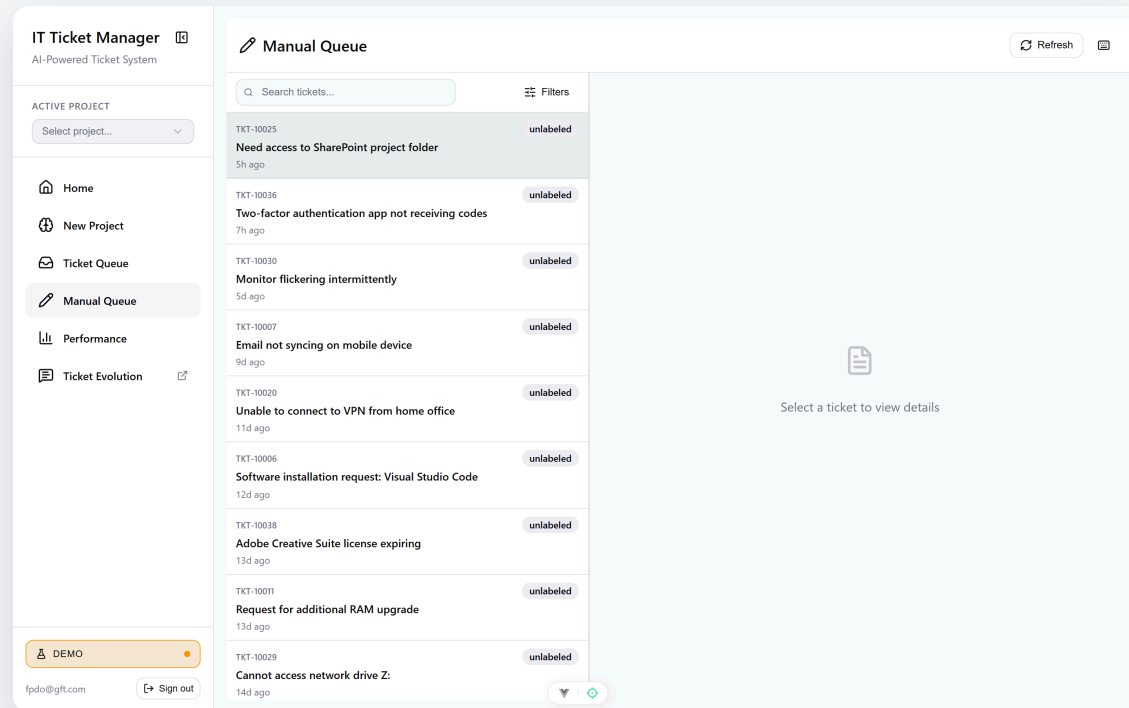


↻ **repeats — every click makes the model better**

Every click is feedback. The model retraines, grows more confident, and reaches target accuracy with **fewer labels** than random sampling — human effort goes only where the model is unsure.

Step 5 · Manual Queue

Bootstrap & ground truth



Flow: The same queue **without AI** — pick the team from a dropdown and **Confirm**. Used to gather the first labels before the model is trained, or to check the model against unaided human judgment.

Step 6 · Ticket Evolution

An AI-drafted first reply — *Resolution*

The screenshot shows the 'Ticket Evolution' interface. At the top, there's a 'No team prediction' dropdown and a '5' value. Below this, a message says 'Suggests the team with a trained AI before resolving.' There are two dropdowns for 'Service category (optional)' and 'Service subcategory (optional)', both with 'Select category...' and 'Select subcategory...' text. A 'Clear' button and a 'Generate solution' button are present. Below this is a 'Category' section with a 'CATEGORY' dropdown showing 'vpn_request', a 'SUGGESTED TEAM' dropdown showing '(GI-UX) Network Access', a 'CERTAINTY' progress bar at 72.0%, and a 'QUALITY CHECK' button with 'No prior feedback'. Below this is a 'Suggested Response' section with a text area containing a draft reply: 'Hi, Thanks for reporting the issue with "VPN connection keeps dropping every few minutes". Please reinstall the VPN client from the Software Center and reboot. If the connection still drops, switch your Wi-Fi to the 5GHz band and disable adapter power-saving. Let me know if the problem continues and we will investigate further. Best regards'. There are 'Copy', 'Regen', and 'Send to Team' buttons. At the bottom is a 'Similar Past Replies (3)' section with a message 'Rate a reply (👍 / 👎) or reuse it as your response' and a list of replies, including one with ID 'R-531802' and a '70% match'.

1. Retrieve similar past tickets
2. Rank by meaning + category + quality
3. Draft a reply
4. Human approves / edits

Here: `vpn_request` → *Network Access* (72%) with a ready-to-send reply.

Each approved answer instantly grows the knowledge base.

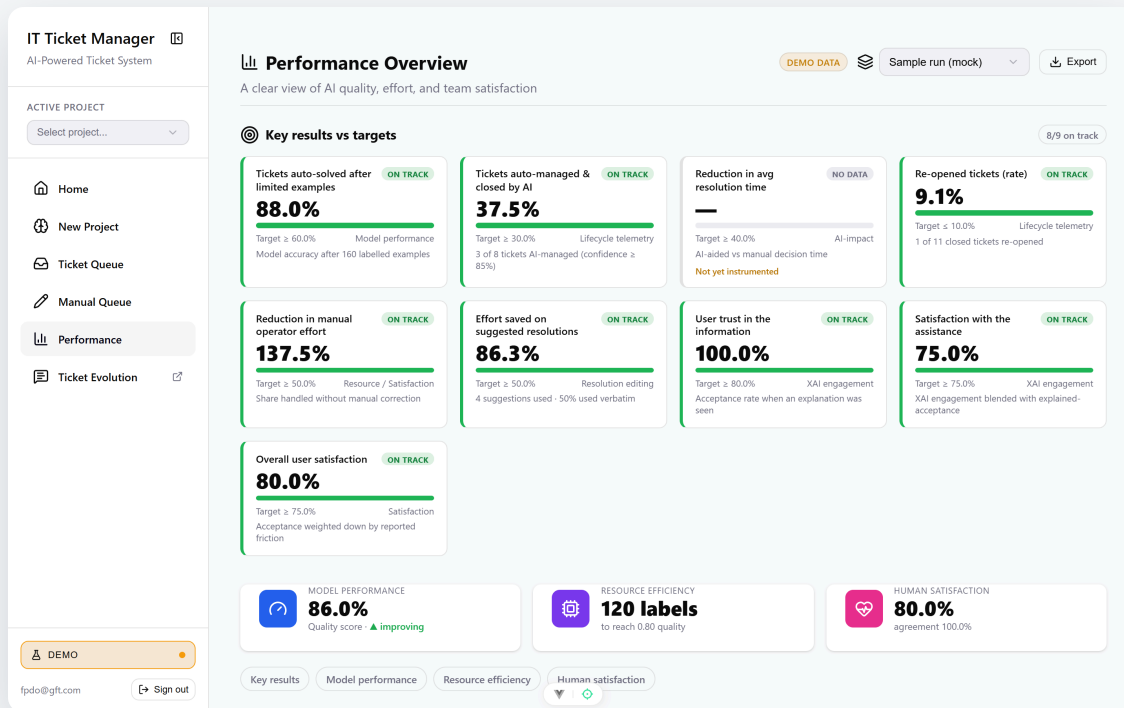
Step 6 · Ticket Evolution — the controls

From ticket to sent reply

Control	What it does
Title · Description	The incoming ticket (<i>required to generate</i>)
AI model (optional)	Predict the team with a trained model first
Suggestions to retrieve (1–20)	How many similar past replies to pull
Service category / subcategory	Sharpen retrieval to the right area
Generate solution	Classify + draft a reply + fetch precedents
Certainty · Quality check	Team confidence · whether community feedback was applied
Suggested Response → Copy · Regen · Send to Team · Export	Edit & use the draft; <i>Send to Team</i> saves it to the KB
Similar Past Replies → 👍 / 👎 · Use this reply	Rate a precedent or reuse it verbatim
Pick a ticket (drawer)	Pull a ticket you already labeled in the queue

Step 7 · Performance

Proving it works — *Benchmarking*



Four pillars, live:

- Model performance — F1, uncertainty
- Resource efficiency — labels/hour, samples-to-target
- Human satisfaction — accept vs. override
- Programme KPIs vs. targets

Step 7 · Performance — the KPIs

Reading the scorecards

Each KPI shows a **value**, a **target** and a **status** — **On track** · *At risk* · *Off target* · *No data*.

KPI	What it measures	Target
F1 score	Overall routing accuracy across all teams	≥ 0.80
Auto-solve rate	Tickets fully resolved by AI, no human touch	≥ 60%
AI-managed rate	Tickets handled with AI assistance	≥ 30%
Resolution-time reduction	Faster handling vs. the manual baseline	≥ 40%
Reopen rate	Closed tickets that bounce back (<i>quality guardrail</i>)	≤ 10%
Manual-effort reduction	Human time saved vs. baseline	≥ 50%
Trust / Satisfaction	Operator confidence & sentiment in the AI	≥ 80% / 75%

Three headline pillars back these up: **Model performance** (F1 + trend) · **Resource efficiency** (labels to reach 0.80, decisions/hour) · **Human satisfaction** (acceptance %).

Human–AI collaboration, quantified

- **Acceptance vs. override by confidence** — proves the model learns *when* to be trusted.
- **Decision time: AI-aided vs. manual** — how much explanations speed agents up.
- **Explainability lift** — acceptance is higher when the explanation was viewed.
- **Auto-close** — high-confidence tickets ($\geq 85\%$) resolve without human touch; **reopen rate** is the safety check.

A repeatable **benchmarking suite** replays realistic scenarios to compare strategies and models fairly.

Progress against programme targets

60%

Tickets ****auto-solved**** by AI

–40%

****Resolution time****
reduction

≤10%

****Reopen rate****
ceiling

80%

User ****trust**** in the
AI

Illustrative programme (AFU) targets. Sample benchmark run: F1 0.83 · Accuracy 0.86 · AUROC 0.89.

Live demo — the exact click-path

1. **Login** → **Home**: point at System Status (API, projects, models).
2. **New Project**: pick *AI Model* + *Labeling Strategy* → *Load from Data* → **Create Project**.
3. **Ticket Queue**: select a ticket → read **84%** suggestion → open **Why this suggestion?** → **Confirm** () one, **Reassign** another.
4. **Ticket Evolution** (*new tab*): enter a ticket → **Generate solution** → review reply + precedents → **Send to Team**.
5. **Performance**: walk the scorecards → **Export** a snapshot.

Fallback: if anything misbehaves live, the screenshots in this deck reproduce every step.

Anticipated questions

They ask...	Say...
<i>"Is our data safe?"</i>	Tickets are anonymised (*_anon fields); runs on-prem with MinIO/DuckDB.
<i>"What if the AI is wrong?"</i>	The human confirms/overrides every label; explanations expose bad calls.
<i>"Does it need lots of labeled data?"</i>	No — active learning reaches target accuracy with far fewer labels.
<i>"What model writes the replies?"</i>	Retrieval (RAG) over past tickets + an LLM draft the human always edits.
<i>"How do we know it's working?"</i>	The Performance page + benchmarking suite — accuracy, effort, trust.
<i>"Can it fully automate?"</i>	Only high-confidence tickets auto-close, with reopen-rate as a guardrail.

Business value

Less manual effort

AI triages and drafts;
humans review instead of
starting from zero.

Faster resolutions

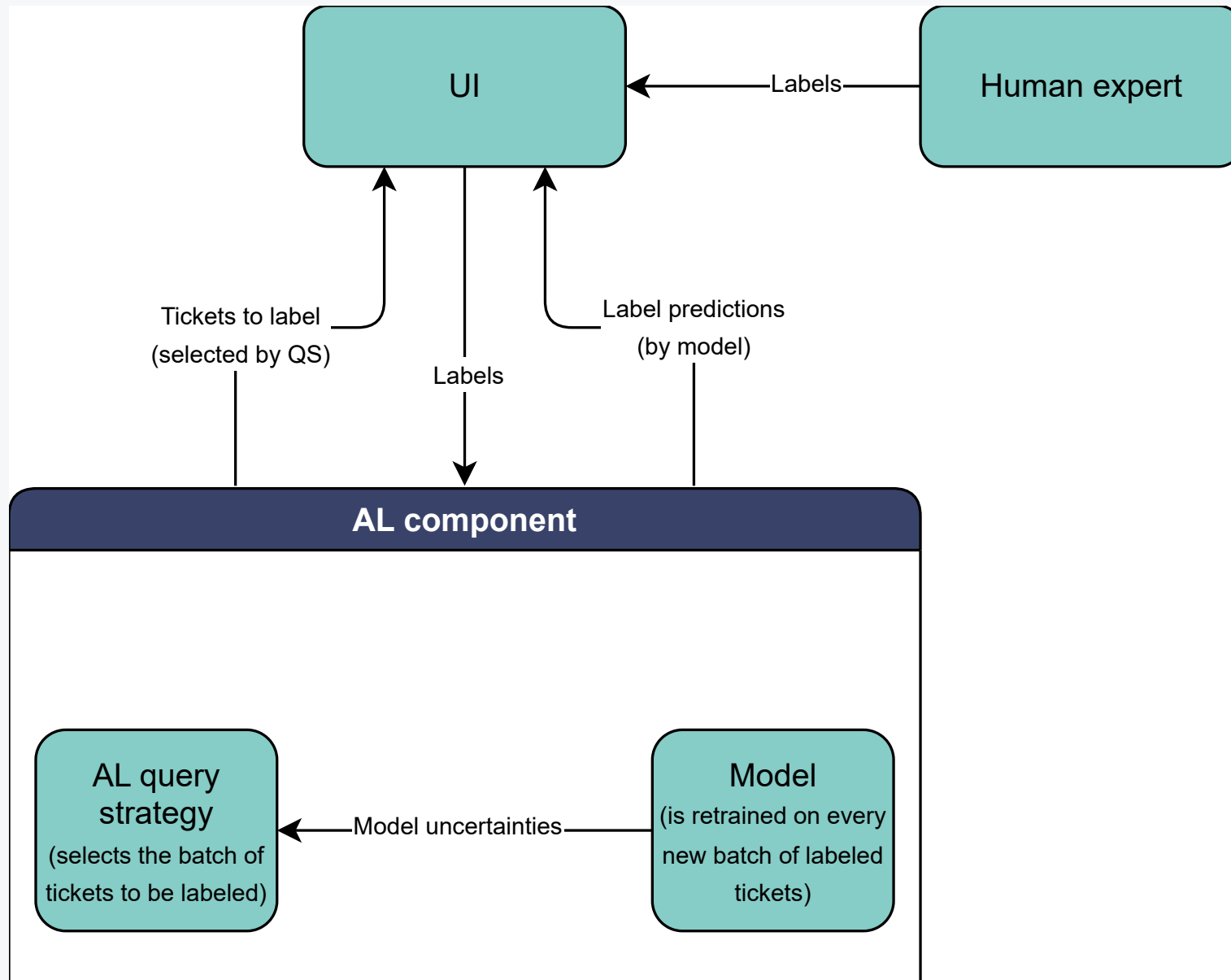
Right team, first time — with
a ready-to-send reply.

Trust by design

Every decision is explained
and every outcome is
measured.

Continuous improvement: the system gets **more capable and more trusted** the more it's used.

Architecture at a glance



Thank you

Key takeaways

- Classify & resolve tickets automatically — with confidence.
- Explain every suggestion — no black box.
- Learn from human feedback — smarter every day.
- Prove the impact — accuracy, effort saved, trust.

Questions & discussion